# Binomial Distribution

## Section 1: Binomial Criteria & Single Probabilities

**1** Which of the following is NOT a required condition for a variable $X$ to have a binomial distribution $B(n, p)$?

**a.** The probability of success changes with each trial

**b.** There is a fixed number of trials $n$

**c.** Each trial results in one of only two outcomes (success or failure)

**d.** Trials are independent of one another

**2** If $X \sim B(6, 0.3)$, calculate $P(X = 3)$ correct to 4 decimal places.

**a.** $0.1852$

**b.** $0.2541$

**c.** $0.3087$

**d.** $0.1176$

**3** If $X \sim B(5, 0.4)$, find $P(X = 0)$.

**a.** $0.0778$

**b.** $0.0102$

**c.** $0.0625$

**d.** $0.1000$

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## Section 2: Cumulative Binomial Probabilities

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**4** If $X \sim B(4, 0.5)$, find $P(X \geq 2)$.

**a.** $\frac{11}{16} = 0.6875$

**b.** $\frac{1}{2} = 0.5000$

**c.** $\frac{5}{16} = 0.3125$

**d.** $\frac{3}{8} = 0.3750$

**5** A manufacturer knows that 5% of lightbulbs are defective. In a random sample of 20 bulbs, what is the probability that at most 1 is defective?

**a.** 0.7358

**b.** 0.6824

**c.** 0.8245

**d.** 0.9128

**6** A fair coin is tossed 6 times. Find the probability of obtaining at least 5 heads.

**a.** $\frac{7}{64} \approx 0.1094$

**b.** $\frac{6}{64} \approx 0.0938$

**c.** $\frac{1}{64} \approx 0.0156$

**d.** $\frac{11}{64} \approx 0.1719$

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## Section 3: Mean & Variance of the Binomial Distribution

**7** A random variable $X \sim B(48, 0.25)$. Find its mean and standard deviation.

**a.** Mean $= 12, \sigma = 3$

**b.** Mean $= 12, \sigma = 9$

**c.** Mean $= 16, \sigma = 4$

**d.** Mean $= 12, \sigma = \sqrt{3}$

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**8** For a binomial distribution $X \sim B(n, p)$, the mean is 6 and the variance is 3.6. Find $n$ and $p$.

**a.** $n = 15, p = 0.4$

**b.** $n = 20, p = 0.3$

**c.** $n = 12, p = 0.5$

**d.** $n = 24, p = 0.25$

**9** What is the maximum possible variance of a binomial distribution with $n = 40$ trials?

**a.** 10

**b.** 20

**c.** 5

**d.** 40

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## Section 4: Contextual Modeling & Problem Solving

**10** A treatment is successful for 80% of patients. If 10 patients are treated, what is the most likely number of successful outcomes (the mode)?

**a.** 8

**b.** 7

**c.** 9

**d.** 10

**11** How many times must a fair die be rolled so that the probability of getting at least one 6 is greater than $0.9$ ?

**a.** At least 13 rolls

**b.** At least 10 rolls

**c.** At least 6 rolls

**d.** At least 18 rolls

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**12** In an exam, 70% of candidates pass. Out of 8 randomly chosen candidates, what is the probability that exactly 6 pass, given that at least 5 pass?

**a.** $\approx 0.380$

**b.** $\approx 0.520$

**c.** $\approx 0.296$

**d.** $\approx 0.420$

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